CONDITIONS ON AN ALGEBRAIC SPACE OVER A FIELD IN TERMS OF ITS CLOSED POINTS

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Conditions on points over fields. lemma-conditions-on-point-on-space-over-field (carded L26742). The pointwise conditions.
- Closed points of schemes over fields. Morphisms, lemma closed-points-generic-points (cross-chapter, named). Closed points of the presentation.

Source reference. Original reference 02NG: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01T6: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0397: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AAX: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NM: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NW: focused-seeds/geometry/S9-decent-conditions-on-space-over-field/ref-tags.txt, source lines 10–10. Navigation pointer

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be an algebraic space locally of finite type over a field k. Consider the conditions*

(1) *$|X|$ is a finite set,*
(2) *$|X|$ is a discrete space,*
(3) $\dim(|X|) = 0$,
(4) $\dim(X) = 0$,
(5) *$X \to \mathrm{Spec}(k)$ is locally quasi-finite,*

Then (2), (3), (4), and (5) are equivalent. If X is decent, then (1) implies the others.

Proof. Parts (4) and (5) are equivalent for example by Morphisms of Spaces, Lemma 04NW.

Let $U \to X$ be a surjective étale morphism where U is a scheme.

If $\dim(U) > 0$, then choose a nontrivial specialization $u \rightsquigarrow u'$ in U and the transcendence degree of $\kappa(u)$ over k exceeds the transcendence degree of $\kappa(u')$ over k. It follows that the images x and x' in X are distinct, because the transcendence degree of x/k and x'/k is well defined, see Morphisms of Spaces, Definition 04NM. We conclude that (2) and (3) imply (4).

Conversely, if $X \to \mathrm{Spec}(k)$ is locally quasi-finite, then U is locally Noetherian (Morphisms, Lemma 01T6) of dimension 0 (Morphisms, Lemma 0397) and hence is a disjoint union of spectra of Artinian local rings (Properties, Lemma 0AAX). Hence U is a discrete topological space, and since $|U| \to |X|$ is continuous and open, the same is true for $|X|$. In other words, (4) implies (2) and (3).

Assume X is decent and (1) holds. Then we may choose U above to be affine. The fibres of $|U| \to |X|$ are finite (this is a part of the defining property of decent spaces). Hence U is a finite type scheme over k with finitely many points. Hence U is quasi-finite over k (Morphisms, Lemma 02NG) which by definition means that $X \to \mathrm{Spec}(k)$ is locally quasi-finite. □